Research report

Audience and scope

First-year CSE students with no programming experience. Assumed access: a computer and Python or Google Colab. Daily learning time is 30 minutes for 28 days, totaling 14 hours. This is an introduction, not evidence of professional proficiency. Installation may require separate support time.

Evidence summary

1. Python Software Foundation, The Python Tutorial: https://docs.python.org/3/tutorial/ . The tutorial explicitly targets programmers new to Python, not absolute programming beginners. Therefore this pathway adds trace tables, small tasks and vocabulary explanations rather than assigning the entire tutorial unsupported.
2. Python introduction: https://docs.python.org/3/tutorial/introduction.html . Reference for numbers, strings and lists.
3. Python control flow: https://docs.python.org/3/tutorial/controlflow.html . Reference for conditions, loops, range and functions.
4. Python data structures: https://docs.python.org/3/tutorial/datastructures.html . Reference for list operations and dictionaries.
5. Python errors: https://docs.python.org/3/tutorial/errors.html . Reference for exception handling. This specific page is supplied as a further-reading reference, not represented as individually read during preparation.
6. Dunlosky et al. (2013), Improving Students' Learning With Effective Learning Techniques. Association for Psychological Science summary: https://www.psychologicalscience.org/journals/pspi/1529100612453266/ . Practice testing and distributed practice received high utility ratings. This supports daily retrieval and repeated review, but does not prove this exact 30-minute Python intervention is effective.
7. Qian, Y. and Lehman, J. (2017), Students' Misconceptions and Other Difficulties in Introductory Programming: A Literature Review, ACM Transactions on Computing Education 18(1), Article 1. DOI: 10.1145/3077618. Discovery record: https://scispace.com/papers/students-misconceptions-and-other-difficulties-in-292qppb2ot . The available abstract describes syntactic, conceptual and strategic difficulties. Full-text assessment of the original review was not performed; do not claim otherwise.

Design decisions

Use five minutes of retrieval, seven minutes of a worked example, thirteen minutes of independent coding and five minutes of testing/reflection. Week 1 covers values and input; Week 2 decisions and loops; Week 3 collections and functions; Week 4 validation and integration. Revisit earlier content during later exercises. Retrieval means attempting an answer without looking, not simply rereading.

Distinguish syntax errors (missing colon), conceptual errors (thinking input is a number) and strategic errors (not breaking down a problem). Trace variable values on paper before running code. Fade help from a complete example to a partial example to an independent task.

Patient AI tutoring protocol

Ask what the learner predicts and what they tried. Give one hint, not the whole answer. Ask the learner to run a minimal example and report the exact error without personal information. Explain one concept in plain language. Ask for an explanation in the learner's own words and a new test case. Release a full solution only after a genuine attempt or an explicit request. Treat generated code as untrusted until reviewed and run.

Research workflow still required

Use two distinct AI research tools, such as Perplexity plus Elicit or Consensus. Search for novice programming misconceptions and beginner practice strategies. Record exact query, tool, date, source URL, supported claim and limitation in tool_evidence.md. Search results here are not evidence that Elicit or Consensus was used. Check each paper's actual bibliographic details and abstract before citing it in a final academic report.

Evaluation

Administer the quiz before the pathway and after day 28 without answers or AI help. Use the same scoring convention and disclose repeated-test familiarity. Record exercise correctness, hint count and learner explanations. A one-person pre/post comparison cannot establish causation. No measured learner improvement is claimed.

Limitations

Fourteen hours cannot cover all Python topics. File operations, classes, networking and package management are intentionally out of scope. Public documentation can change; this project uses basic Python 3 syntax and records the actual execution version separately. Tool free-tier access must be checked by the student; no paid features are required by the supplied code.